

Phase 2 — Model Design Specification

Aggregating User Preferences while Ensuring Equity, Diversity, and Inclusion
using Graph Summarization

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1. Graph-Based Preference Model

1.1 Formal Definition

The preference data is represented as a **weighted attributed bipartite graph**:

$$G = (U \cup I, E, w, s)$$

Symbol	Definition
$U = \{u_1, u_2, \dots, u_m\}$	Set of user nodes
$I = \{i_1, i_2, \dots, i_n\}$	Set of item nodes (e.g., movies in MovieLens)
$E \subseteq U \times I$	Set of edges — $(u, i) \in E$ iff user u rated item i
$w : E \rightarrow [0.5, 5]$	Weight function — $w(u, i)$ = rating given by u to i
$s : U \rightarrow \{F, M\}$	Sensitive attribute — gender of each user (Female / Male)

Key property (bipartite): No edges exist between two users or two items. Only user–item edges exist.

1.2 Sensitive Attribute Choice

Chosen attribute: gender (user side)

- Directly tied to the research question: does the collective ranking fairly represent female and male users?
- Gender is a clean binary field in MovieLens (100k, 1M), requiring no pre-processing
- Reference metrics (Yao & Huang 2017; Leonhardt et al. 2018) use gender on users → direct comparability with literature
- **Limitation to acknowledge in the paper:** binary F/M encoding does not reflect gender diversity in reality; this is a methodological starting point to be extended

1.3 Toy Example

4 users, 3 items, gender attribute:

User	Gender	i_1	i_2	i_3
u_1	F	5	2	—
u_2	F	4	—	1
u_3	M	—	5	4
u_4	M	1	—	5

Groups: $G_F = \{u_1, u_2\}$, $G_M = \{u_3, u_4\}$

2. From Graph to Collective Ranking

2.1 Paradigm Choice

EDI metrics → anchored in ratings (raw values)

- MovieLens provides raw ratings; no conversion needed
- Preserves preference intensity ("love" $\neq$ "like")
- Directly matches Yao & Huang's metrics

Baselines → use ranks

- Borda and Condorcet are rank-based by definition
- Allows comparison between "rating aggregation" vs "rank aggregation" — an interesting experimental result in itself

2.2 Baseline Aggregation Methods

Method	Formula	Known EDI problem
Average score	$\text{score}(i) = (1/ U_i) \times \sum_{u:(u,i) \in E} w(u,i)$	Favors items popular within the majority group
Borda	$\text{score}_B(i) = \sum_u \text{rank}_u(i)$	Biased when group sizes are unequal
Condorcet	$i > j \text{ if } \{u : w(u,i) > w(u,j)\} > U /2$	Majority can systematically override minority

→ None of these methods explicitly controls EDI during aggregation — this gap motivates the proposed model.

2.3 Toy Example — Aggregation

Average score:

- i_1 : $(5 + 4 + 1) / 3 = 3.33$
- i_2 : $(2 + 5) / 2 = 3.50$
- i_3 : $(1 + 4 + 5) / 3 = 3.33$

Collective ranking: $i_2 > i_1 \approx i_3$

- G_F prefers: $i_1 > i_2 > i_3$
- G_M prefers: $i_3 > i_2$ (i_1 rated low)
- Global: i_2 first — appears neutral, but **i_1 (strongly preferred by women) is demoted**

→ The global ranking is not equitable: women's preferences are underrepresented.

3. Formal EDI Metrics

Let $R = (i_1, \dots, i_k)$ be the collective top-k ranking. Let G_F, G_M be the two user groups.

3.1 E — Equity (fairness between user groups)

$$\text{utility}_g(R) = (1/|G_g|) \times \sum_{u \in G_g} \sum_{i \in R} w(u, i) / k$$

$$\Delta E = |\text{utility}_F(R) - \text{utility}_M(R)|$$

- **Interpretation:** Average satisfaction of each group with the collective top-k
- **Target:** $\Delta E \approx 0$ (both groups equally satisfied)
- **Source:** Yao & Huang (2017) — value unfairness metric

3.2 D — Diversity (variety of recommended items)

$$\text{ILD}(R) = (2 / k(k-1)) \times \sum_{i \neq j \in R} \text{dist}(i, j)$$

$$\text{dist}(i, j) = 1 - \text{cosine_similarity}(v_i, v_j)$$

where $v_i \in \mathbb{R}^m$ is the rating vector of item i across all users (0 if unrated).

- **Interpretation:** Average pairwise dissimilarity within the top-k list
- **Target:** ILD high (varied recommendations, not all the same genre)
- **Parameter to justify:** cosine similarity on rating vectors — alternative: genre-based distance

3.3 I — Inclusion (minority representation in results)

$$\text{inclusion}_g(R) = (1/|G_g|) \times \sum_{u \in G_g} |\{i \in R : w(u, i) \geq \theta\}| / k$$

θ = minimum rating considered "relevant" (e.g., $\theta = 4.0$)

$\alpha_g = |G_g| / |U|$ (population share of group g)

- **Interpretation:** Fraction of top-k items that group g members find relevant
- **Target (proportional fairness — Kellerhals & Peters 2024):**

$$\text{inclusion}_g(R) \geq \alpha_g$$

If women represent 30% of users, at least 30% of top-k items should be relevant to them.
Grounded in the "core" concept: a group of size α_g deserves at least α_g of the collective outcome.

- **Parameter to justify:** choice of θ ($\theta \in \{3.5, 4.0\}$ will be compared experimentally)

4. Proposed Contribution: EDI-Constrained Graph Summarization

4.1 Principle

During graph coarsening (Phase 3), instead of merging nodes freely to minimize structural error, add an **EDI preservation constraint**:

Do not merge two supernodes s_a, s_b if their fusion causes:

- ΔE to increase beyond ϵ_E , **OR**
- $ILD(R)$ to decrease beyond ϵ_D , **OR**
- $\text{inclusion}_F(R)$ to decrease beyond ϵ_I

4.2 Formal Objective (to be refined in Phase 3)

```
min structural_loss(G, G')           [standard coarsening objective]
s.t.  $\Delta E(G') \leq \Delta E(G) + \epsilon_E$ 
       $ILD(G') \geq ILD(G) - \epsilon_D$ 
       $\text{inclusion}_F(G') \geq \text{inclusion}_F(G) - \epsilon_I$ 
```

where G' is the summarized graph and $\epsilon_E, \epsilon_D, \epsilon_I$ are EDI degradation tolerances.

4.3 Research Question (candidate formulation)

Given a weighted bipartite preference graph $G = (U \cup I, E, w, s)$, how can we construct a summary G' (via community detection and coarsening) that reduces size while bounding the degradation of equity (ΔE), diversity (ILD), and inclusion metrics, compared to classical aggregation rules (Borda, Condorcet, average score)?

5. Next Steps

Phase	Action	Deliverable
Phase 2 (June)	Validate model on toy example in Python/NetworkX	Verified model + metrics code
Phase 3 (July)	Design EDI-constrained coarsening algorithm	Algorithm description + complexity analysis
Phase 4 (August)	Implement on MovieLens 100k	Functional prototype + baseline methods
Phase 5 (Sept.)	Compare ΔE , ILD, inclusion vs baselines	Experimental results + statistical analysis

6. Open Parameters (to document in paper)

Parameter	Role	Justification needed
θ	Relevance threshold for inclusion metric	Compare $\theta \in \{3.5, 4.0\}$ experimentally
k	Top-k size	Standard values: $k \in \{5, 10, 20\}$
$\epsilon_E, \epsilon_D, \epsilon_I$	EDI degradation tolerances	Set via baseline measurement on MovieLens
$\text{dist}(i,j)$	Item distance for ILD	Cosine on rating vectors vs genre-based distance